LABORATORY EQUIPMENT BOOKING & MAINTENANCE SYSTEM

Design and Implementation of a Database Management System

Subject
DBMS

Review
1

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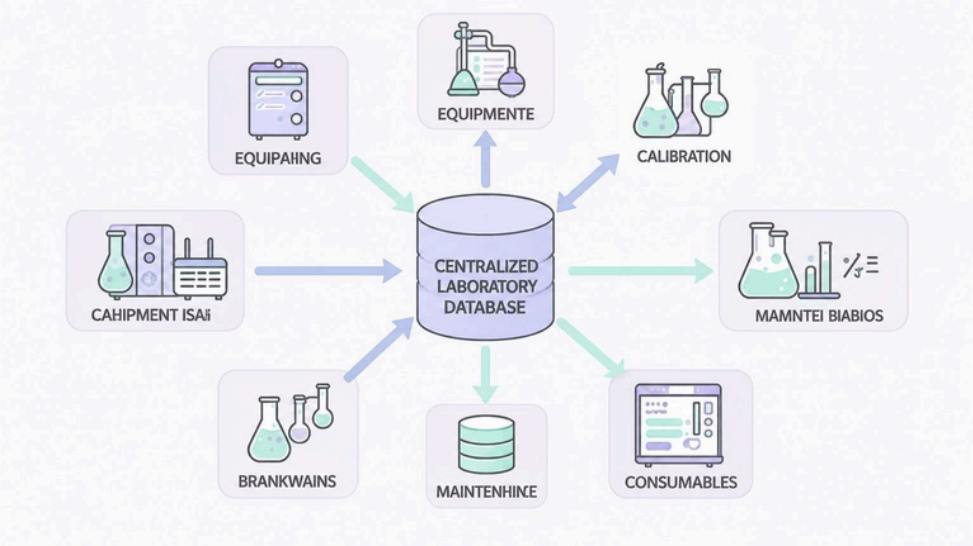
Division
CSE Panthers



PROBLEM STATEMENT

Manual, disconnected tools lead to double bookings, missing history, and unclear equipment status at booking time.

- Double bookings in busy weeks
- Incomplete history (who, when, why) makes tracking hard
- Calibration and maintenance status isn't shown when booking
- Separate logs for breakdowns and consumables cause delays and stockouts



Processes converge to a centralized laboratory database to eliminate fragmentation

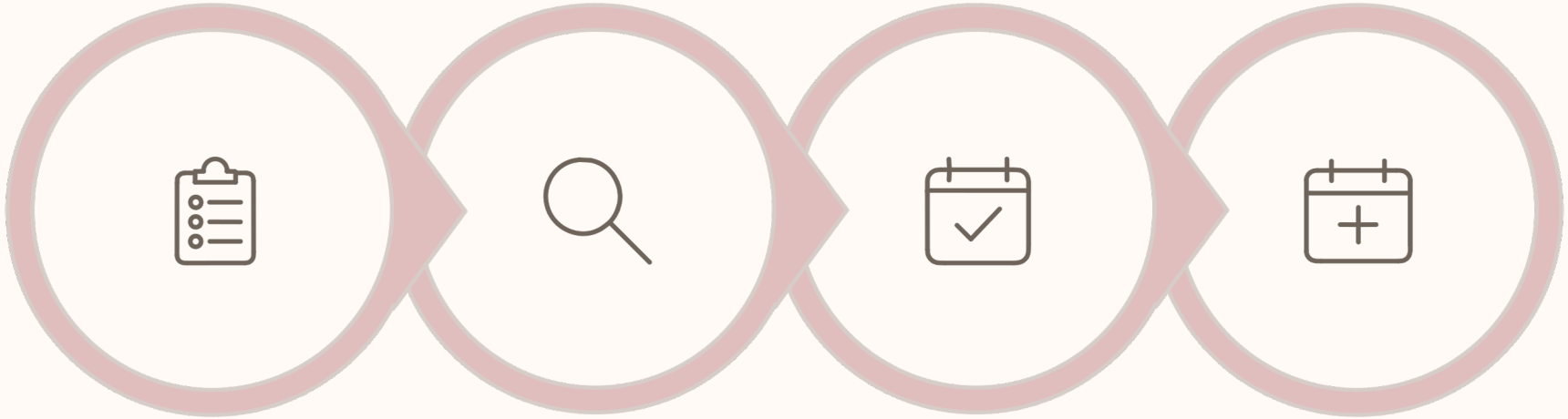
EXISTING SYSTEMS

Disparate tools handle parts of the workflow but lack a single source of truth or dependable availability checks.

EXISTING APPROACH	MAIN USE	OBSERVATION
Manual Registers	Equipment bookings and records	Handwritten or incomplete entries hinder traceability and often cause double bookings or missing history.
Spreadsheets and separate maintenance logs	Equipment and booking data, and maintenance tracking	Version drift and siloed logs lead to outdated availability, duplicate bookings, and missed calibration or repair status.

PROPOSED SOLUTION

A centralized relational database system for managing laboratory equipment from booking and usage to calibration, breakdown and maintenance.



Register **Search** **Check** **Book**

EQUIPMENT → BOOKING → USAGE → CALIBRATION / BREAKDOWN → MAINTENANCE → REPORTS

DISTINGUISHING FEATURES



CONFLICT -AWARE EQUIPMENT BOOKING

Prevents overlapping approved bookings for the same equipment.



EQUIPMENT LIFECYCLE TRACKING

Connects booking, usage, calibration, breakdowns, and maintenance into one history.



CALIBRATION AND MAINTENANCE TRACKING

Maintains calibration records and maintenance logs for full provenance.



AVAILABILITY AND UTILIZATION REPORTING

Reports on availability, utilization, calibration due dates, and downtime for planning and audits.

SYSTEM REQUIREMENTS

Core capabilities required for the Laboratory Equipment Booking and Maintenance System.

1. EQUIPMENT MANAGEMENT

- Manage laboratories
- Manage equipment categories
- Manage equipment
- Track equipment status

2. BOOKING MANAGEMENT

- Search equipment
- Check equipment availability
- Create bookings
- Approve bookings
- Prevent overlapping approved bookings

3. USAGE & CALIBRATION

- Record equipment usage
- Maintain usage logs
- Record calibration information
- Track calibration due dates

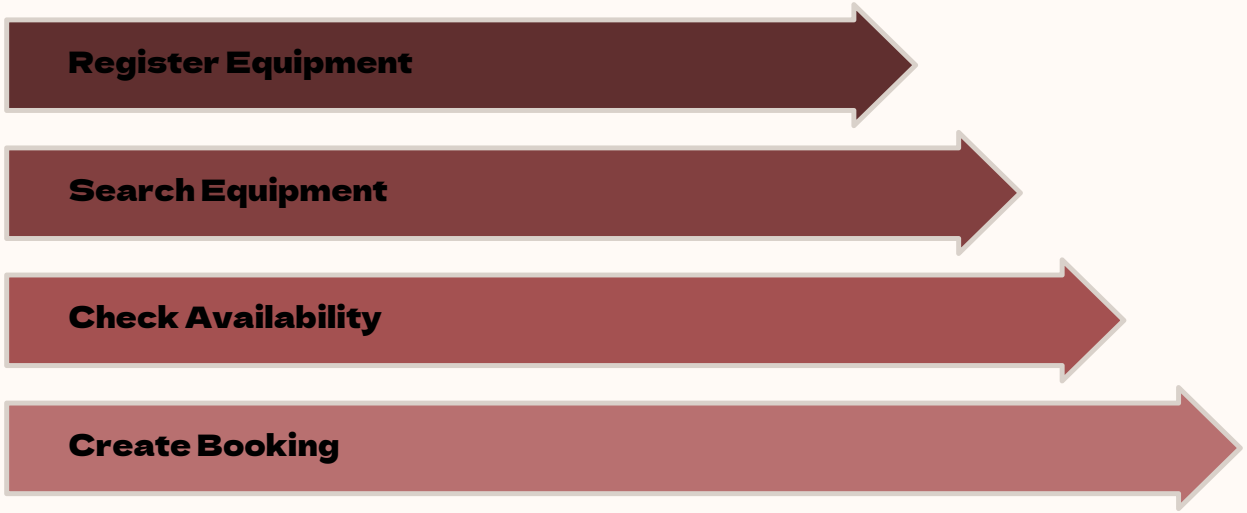
4. MAINTENANCE & INVENTORY

- Report equipment breakdowns
- Manage maintenance jobs
- Manage consumables
- Track stock limits
- Maintain vendor information

5. REPORTING

- Equipment availability
- Equipment utilization
- Calibration due
- Downtime
- Breakdown information
- Vendor service

SYSTEM WORKFLOW AND BUSINESS RULES

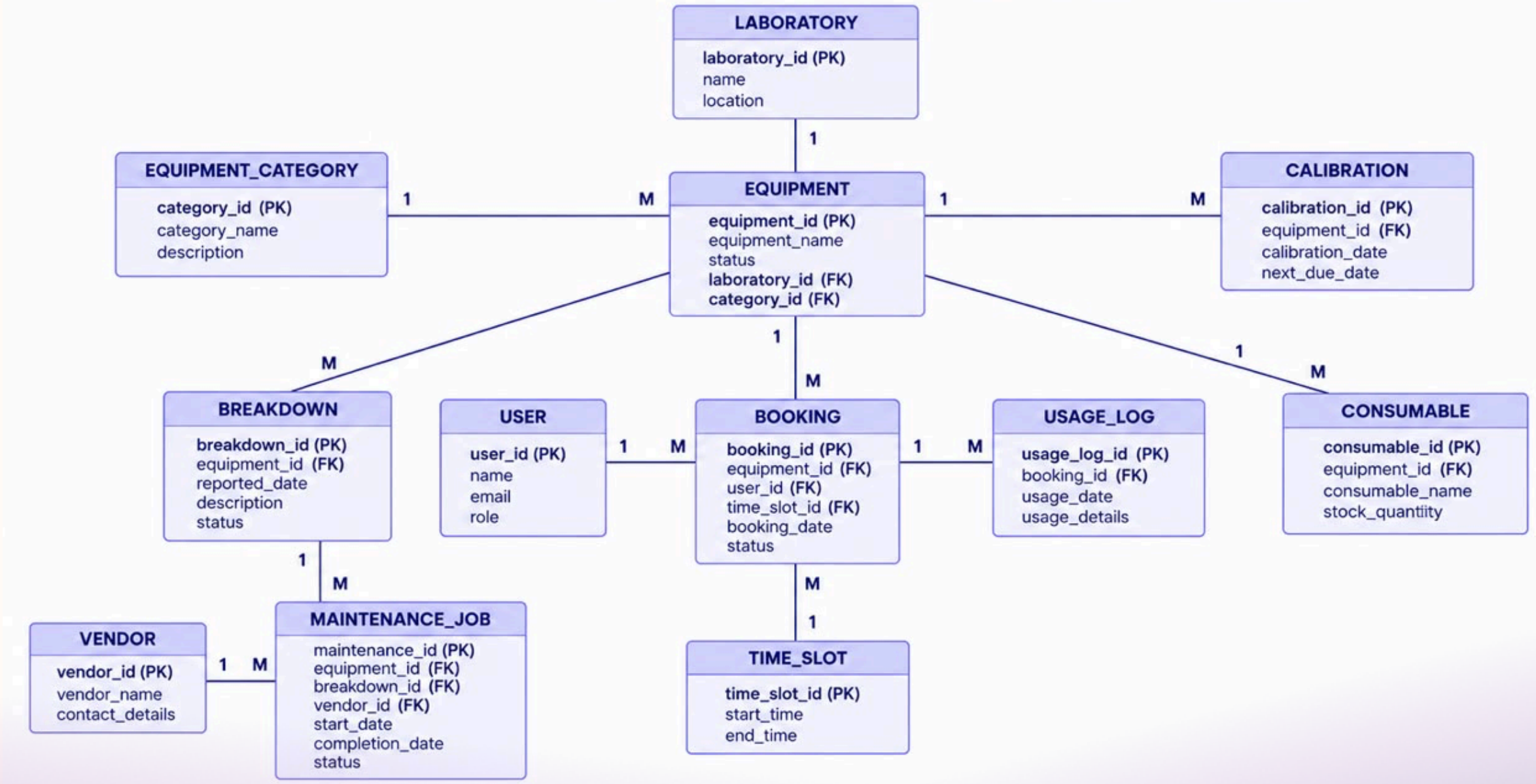


KEY BUSINESS RULES

1. No overlapping approved bookings
2. Only active equipment can be reserved
3. Calibration dates must be valid
4. Consumables must respect stock limits
5. Record maintenance status and completion

Availability checks and status tracking preserve traceability.

ENTITY-RELATIONSHIP DIAGRAM



INITIAL RELATIONAL SCHEMA

LABORATORY EQUIPMENT BOOKING AND MAINTENANCE SYSTEM

LABORATORY
- laboratory_id (PK)
- name
- location

EQUIPMENT_CATEGORY
- category_id (PK)
- category_name
- description

EQUIPMENT
- equipment_id (PK)
- laboratory_id (FK)
- category_id (FK)
- equipment_name
- status

USER
- user_id (PK)
- name
- email
- role

TIME_SLOT
- time_slot_id (PK)
- start_time
- end_time

BOOKING
- booking_id (PK)
- equipment_id (FK)
- user_id (FK)
- time_slot_id (FK)
- booking_date
- status

USAGE_LOG
- usage_log_id (PK)
- booking_id (FK)
- usage_date
- usage_details

CONSUMABLE
- consumable_id (PK)
- equipment_id (FK)
- consumable_name
- stock_quantity

CALIBRATION
- calibration_id (PK)
- equipment_id (FK)
- calibration_date
- next_due_date

BREAKDOWN
- breakdown_id (PK)
- equipment_id (FK)
- reported_date
- description
- status

MAINTENANCE_JOB
- maintenance_id (PK)
- equipment_id (FK)
- breakdown_id (FK)
- vendor_id (FK)
- start_date
- completion_date
- status

VENDOR
- vendor_id (PK)
- vendor_name
- contact_details

PK = Primary Key
FK = Foreign Key

PROJECT PLAN

Three-phaseroadmapalignedtoreviewmilestonesandfinal delivery

PHASE 1 – REVIEW 1

- Requirements gathering
- Use cases and constraints
- ER schema draft

PHASE 2 – REVIEW 2

- Normalize the design
- Define tables and keys
- Implement SQL schema

PHASE 3 – REVIEW 3

- Prototype build
- Sample data and workflows
- Testing and final handoff